

Node Structure Visualization

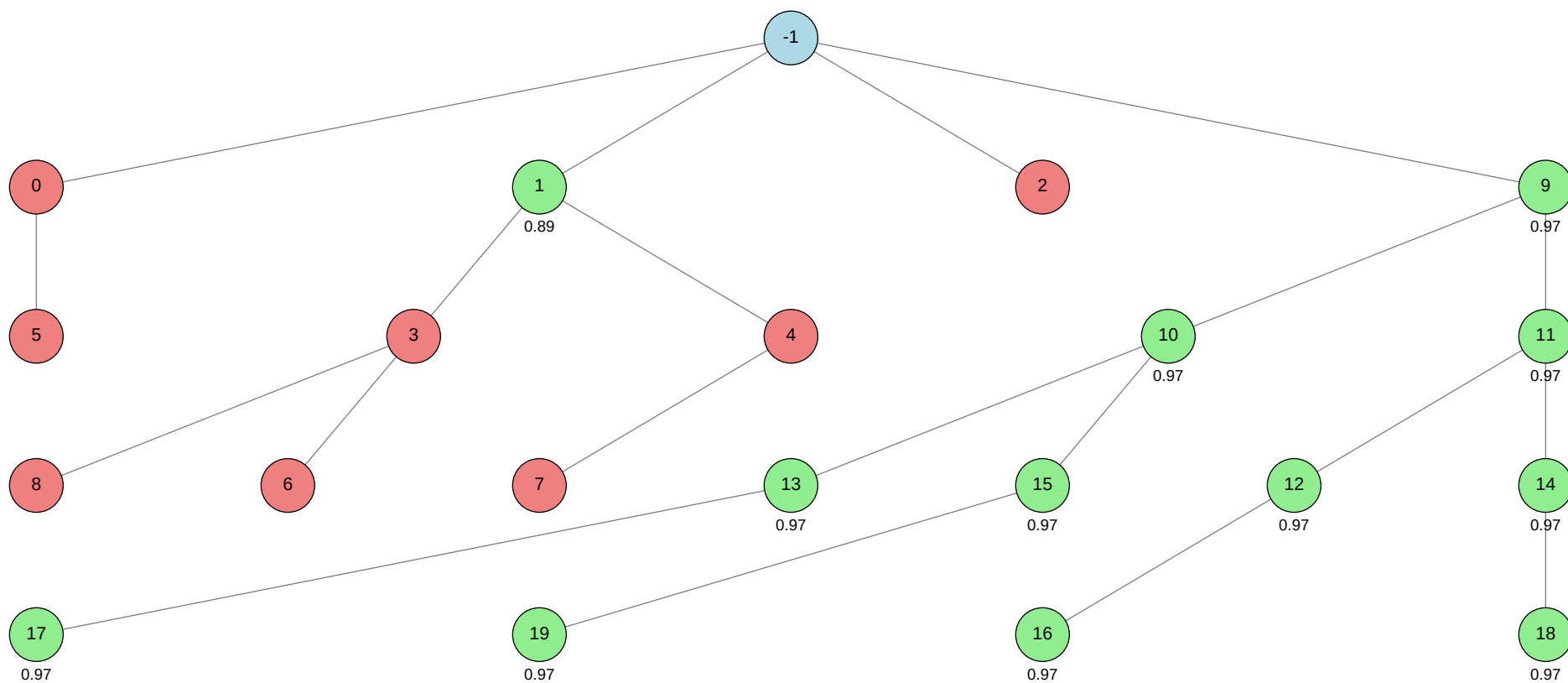
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9720 (Node 11)

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
-1			
0			autogluon.multimodal
1			machine learning
2			autogluon.tabular
3			machine learning
4			machine learning
5			autogluon.multimodal
6			machine learning
7			machine learning
8			machine learning
9			autogluon.multimodal
10			autogluon.multimodal
11			autogluon.multimodal
12			autogluon.multimodal
13			autogluon.multimodal
14			autogluon.multimodal
15			autogluon.multimodal
16			autogluon.multimodal
17			autogluon.multimodal
18			autogluon.multimodal
19			autogluon.multimodal



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	0.3936
ctime	2026-05-20 17:52:09
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	8
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	4
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	11.569199999999997
validated_visits	12
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 9]

Node 0 (evolve)

Property	Value
uct_value	1.7306
ctime	2026-05-20 17:53:17
debug_attempts	1
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	2
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	2
parent_id	-1
child_ids	[5]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 42-addition_2/node_0/conda_env

Resolved 235 packages in 1.62s

Uninstalled 1 package in 1.95s

Installed 233 packages in 1m 00s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260520

+ autogluon-common==1.5.1b20260520

+ autogluon-core==1.5.1b20260520

+ autogluon-features==1.5.1b20260520

+ autogluon-multimodal==1.5.1b20260520

+ autogluon-tabular==1.5.1b20260520

+ autogluon-timeseries==1.5.1b20260520

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.11

+ botocore==1.43.11

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2
- + hyperopt==0.2.7

+ idna==3.15
... [truncated, 6067 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the `MultiModalPredictor.fit()` method in AutoGluon does not accept a `verbose` keyword argument. Passing `verbose=True` causes a `TypeError`, preventing the model from training.

SUGGESTED_FIX: Remove the `verbose=True` argument from the `predictor.fit()` call. Rerun the script after this change to allow the model to train without encountering the unexpected keyword argument error.

Node 1 (evolve)

Property	Value
uct_value	0.7491
avg_raw_score	0.888
ctime	2026-05-20 18:01:52
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.25
exploration	0.7429102097403913
failure_penalty	-0.25
failure_visits	5
id	1
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	3
normalized_score	0.0
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.888

validated_visits	1
validated_weight	0.16666666666666666
validation_score	0.8880
visits	6
parent_id	-1
child_ids	[3, 4]

Node 2 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-20 18:19:03
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2131672923786587
failure_penalty	-0.0
failure_visits	1
id	2
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 42-addition_2/node_2/conda_env

Resolved 239 packages in 1.35s

Uninstalled 1 package in 2.23s

Installed 237 packages in 2m 21s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260520

+ autogluon-common==1.5.1b20260520

+ autogluon-core==1.5.1b20260520

+ autogluon-features==1.5.1b20260520

+ autogluon-multimodal==1.5.1b20260520

+ autogluon-tabular==1.5.1b20260520

+ autogluon-timeseries==1.5.1b20260520

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.11

+ botocore==1.43.11

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2

+ hyperopt==0.... [truncated, 24331 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon Tabular failed to train any models because all candidate models were skipped with the message "No valid features to train ...", resulting in a RuntimeError. This happened because the tabular features provided (after preprocessing) were insufficient—likely only 'image_name' was present, which is not a usable feature for modeling, so no valid input features remained for training.

SUGGESTED_FIX:

Check your feature selection and preprocessing steps to ensure that at least one meaningful, non-identifier feature (e.g., numeric or categorical columns other than 'image_name') is included in both train and test sets. If you intend to use image data, switch to AutoGluon Multimodal, properly specify the image path column, and configure feature metadata so the model can use image pixels for prediction.

Node 3 (evolve)

Property	Value
uct_value	0.9261
ctime	2026-05-20 18:33:23
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	3
id	3
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	2
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	evolve
time_step	3
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	3
parent_id	1
child_ids	[8, 6]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 42-addition_2/node_3/conda_env
Resolved 79 packages in 1.67s
Uninstalled 1 package in 3.05s
Installed 78 packages in 3m 09s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```


- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.15.0
- + idna==3.15
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.13.4
- + pydantic-core==2.46.4
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 534010 chars total]

Error Analysis:

ERROR_SUMMARY: The model achieves a validation AUROC of 0.5000 on the holdout set, indicating it is not learning to distinguish between malignant and non-neoplastic lesions—its predictions are equivalent to random guessing. This suggests a fundamental issue with either the feature extraction, data preprocessing, label encoding, or model training pipeline.

SUGGESTED_FIX:

Carefully check that the extracted features are meaningful (not all zeros or constant), verify that label encoding is correct and matches the intended classes, and ensure the model is being trained on the correct data splits. Additionally, inspect a few feature vectors and labels manually, and try training a simple baseline (e.g., logistic regression on raw pixel means) to confirm the pipeline is functioning before reintroducing the full feature extraction and LightGBM model.

Node 4 (evolve)

Property	Value
uct_value	1.3384
ctime	2026-05-20 19:00:34
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	2
id	4
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	evolve
time_step	4
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	2
parent_id	1
child_ids	[7]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 42-addition_2/node_4/conda_env
Resolved 79 packages in 3.61s
Uninstalled 1 package in 3.15s
Installed 78 packages in 1m 01s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.15.0
- + idna==3.15
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.13.4
- + pydantic-core==2.46.4
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 534362 chars total]

Error Analysis:

ERROR_SUMMARY: The root cause is that the LightGBM model failed to learn any meaningful decision boundaries: it stopped early with "No further splits with positive gain," and the validation AUROC is 0.5000 (random). This indicates that the extracted features from EfficientNet-B3 are not informative for the malignancy classification task, likely due to issues in feature extraction, data preprocessing, or label encoding.

SUGGESTED_FIX:

Double-check that the image feature extraction pipeline is working correctly (e.g., verify that extracted features are not all zeros or constant, and that images are being loaded and preprocessed as expected). Also, confirm that the label encoding is correct and that the training data is balanced and not corrupted. Try visualizing a few extracted feature vectors and their distributions, and consider training a simple classifier (e.g., logistic regression) to see if any signal exists in the features before using LightGBM.

Node 5 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-20 19:23:55
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	5
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	0
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 42-addition_2/node_5/conda_env

Resolved 235 packages in 5.07s

Uninstalled 1 package in 2.89s

Installed 233 packages in 1m 38s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260520

+ autogluon-common==1.5.1b20260520

+ autogluon-core==1.5.1b20260520

+ autogluon-features==1.5.1b20260520

+ autogluon-multimodal==1.5.1b20260520

+ autogluon-tabular==1.5.1b20260520

+ autogluon-timeseries==1.5.1b20260520

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.11

+ botocore==1.43.11

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15
... [truncated, 9368 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the ``class_weights`` (used as the ``alpha`` parameter for focal loss in AutoGluon) is being passed as a list of strings or an incorrectly formatted list, rather than as a list of floats. The FocalLoss implementation expects ``alpha`` to be a float or a list/array of floats, and passing a list containing string values (e.g., ``['0.7', '0.3']`` instead of ``[0.7, 0.3]``) leads to the `ValueError: "too many dimensions 'str'"`.

SUGGESTED_FIX:

Check how ``class_weights`` is computed and ensure it is a list of floats, not strings. Explicitly convert the weights to floats (e.g., using ``class_weights = [float(w) for w in class_weights]``) before passing them to the ``"optim.focal_loss.alpha"`` hyperparameter in ``predictor.fit``.

Node 6 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-20 19:34:08
debug_attempts	1
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	6
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	debug
time_step	6
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 42-addition_2/node_6/conda_env
Resolved 79 packages in 1.58s
Uninstalled 1 package in 2.48s
Installed 78 packages in 55.67s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.15.0
+ idna==3.15

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 2794 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurred because the Python script attempted to import the `autogluon` package (`from autogluon.multimodal import MultiModalPredictor`), but this package was not installed in the current conda environment. As a result, a `ModuleNotFoundError: No module named 'autogluon'` was raised, causing the script to terminate before any model training or prediction could take place.

SUGGESTED_FIX:

Update your environment setup to explicitly install the `autogluon` package (specifically, `autogluon.multimodal`) in your conda environment before running the script. Add a step in your bash script to install `autogluon.multimodal` via pip (e.g., `pip install autogluon.multimodal`) after activating the environment and before executing the Python script.

Node 7 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-20 19:40:20
debug_attempts	1
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	7
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	7
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 42-addition_2/node_7/conda_env
Resolved 79 packages in 397ms
Uninstalled 1 package in 2.35s
Installed 78 packages in 50.55s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.15.0
+ idna==3.15

```



```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 555260 chars total]
```

Error Analysis:

ERROR_SUMMARY: The script executes without runtime errors, but the trained model achieves a validation AUROC of 0.5000 on the holdout set, indicating it is making predictions equivalent to random guessing. This suggests a critical failure in the feature extraction, data processing, or model training pipeline—most likely, the extracted features are not informative (e.g., all zeros or constant), or there is a mismatch between features and labels.

SUGGESTED_FIX:

First, inspect the extracted features (e.g., print summary statistics for `X_train` and `X_val`) to ensure they are not all zeros or constant values. Next, verify that the image paths are correct and images are being loaded and preprocessed as expected. Finally, confirm that the labels are correctly mapped and that the model is being trained on the correct data; if all checks pass, try training a simple baseline (e.g., logistic regression) to isolate whether the issue is with feature extraction or model training.

Node 8 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-20 20:04:35
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	8
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	8
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 42-addition_2/node_8/conda_env
Resolved 79 packages in 978ms
Uninstalled 1 package in 3.34s
Installed 78 packages in 1m 03s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.15.0
+ idna==3.15

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 533378 chars total]
```

Error Analysis:

ERROR_SUMMARY: The root cause of the poor model performance (AUROC = 0.5000) is that the extracted features from EfficientNet-B3 are not informative for the LightGBM classifier—likely because the feature extraction process is not producing meaningful variation (e.g., all features are nearly constant or zero). This is supported by LightGBM's warning "No further splits with positive gain," which indicates the features do not help separate the classes. Additionally, there is a warning

about the CUDA driver being too old, which may cause the model to run on CPU or lead to improper feature extraction.

SUGGESTED_FIX:

First, verify that the feature extraction step is working correctly by inspecting the extracted feature arrays for variance and non-zero values; ensure images are being loaded and processed as expected. Next, check for any silent errors or device mismatches (e.g., CUDA driver issues causing fallback to CPU or failed model inference). If features are constant or mostly zeros, debug the image loading and preprocessing pipeline, and consider updating the NVIDIA driver or running the feature extraction on CPU to ensure correctness.

Node 9 (debug)

Property	Value
uct_value	1.7262
avg_raw_score	0.9709199999999998
ctime	2026-05-20 20:26:19
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.9871428571428549
exploration	0.7672743653471986
failure_penalty	-0.0
failure_visits	0
id	9
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9871428571428549
num_children	2
prev_tutorial_prompt	None
stage	debug
time_step	9
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.9871428571428549
validated_reward	10.681199999999997

validated_visits	11
validated_weight	1.0
validation_score	0.9666
visits	11
parent_id	-1
child_ids	[10, 11]

Node 10 (evolve)

Property	Value
uct_value	1.9664
avg_raw_score	0.97065
ctime	2026-05-20 21:13:33
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.9839285714285718
exploration	1.0728209804752984
failure_penalty	-0.0
failure_visits	0
id	10
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9839285714285718
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	10
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9839285714285718
validated_reward	4.8546
validated_visits	5
validated_weight	1.0
validation_score	0.9666
visits	5
parent_id	9
child_ids	[13, 15]

Node 11 (evolve)

Property	Value
uct_value	1.9792
avg_raw_score	0.9719999999999999
ctime	2026-05-20 22:12:01
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.9999999999999987
exploration	0.9595602560122969
failure_penalty	-0.0
failure_visits	0
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9999999999999987
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	11
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9999999999999987
validated_reward	4.859999999999999
validated_visits	5
validated_weight	1.0
validation_score	0.9720
visits	5
parent_id	9
child_ids	[12, 14]

Node 12 (evolve)

Property	Value
uct_value	2.2684
avg_raw_score	0.972
ctime	2026-05-20 23:02:46
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	1.0
exploration	1.1772322201769845
failure_penalty	-0.0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	1.0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	12
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	1.0
validated_reward	1.944
validated_visits	2
validated_weight	1.0
validation_score	0.9720
visits	2
parent_id	11
child_ids	[16]

Node 13 (evolve)

Property	Value
uct_value	2.2684
avg_raw_score	0.972
ctime	2026-05-20 23:53:39
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	1.0
exploration	1.1772322201769845
failure_penalty	-0.0
failure_visits	0
id	13
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	1.0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	13
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	1.0
validated_reward	1.944
validated_visits	2
validated_weight	1.0
validation_score	0.9720
visits	2
parent_id	10
child_ids	[17]

Node 14 (evolve)

Property	Value
uct_value	2.2684
avg_raw_score	0.972
ctime	2026-05-21 00:41:47
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	1.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	1.0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	14
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	1.0
validated_reward	1.944
validated_visits	2
validated_weight	1.0
validation_score	0.9720
visits	2
parent_id	11
child_ids	[18]

Node 15 (evolve)

Property	Value
uct_value	2.2684
avg_raw_score	0.972
ctime	2026-05-21 01:27:58
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	1.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	0
id	15
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	1.0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	15
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	1.0
validated_reward	1.944
validated_visits	2
validated_weight	1.0
validation_score	0.9720
visits	2
parent_id	10
child_ids	[19]

Node 16 (evolve)

Property	Value
uct_value	2.1772
ctime	2026-05-21 02:16:36
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	16
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	16
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.972
validated_visits	1
validation_score	0.9720
visits	1
parent_id	12
child_ids	[]

Node 17 (evolve)

Property	Value
uct_value	2.1772
ctime	2026-05-21 02:59:55
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	17
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.972
validated_visits	1
validation_score	0.9720
visits	1
parent_id	13
child_ids	[]

Node 18 (evolve)

Property	Value
uct_value	2.1772
ctime	2026-05-21 03:42:54
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	18
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.972
validated_visits	1
validation_score	0.9720
visits	1
parent_id	14
child_ids	[]

Node 19 (evolve)

Property	Value
uct_value	2.1772
ctime	2026-05-21 04:26:23
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	19
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.972
validated_visits	1
validation_score	0.9720
visits	1
parent_id	15
child_ids	[]

